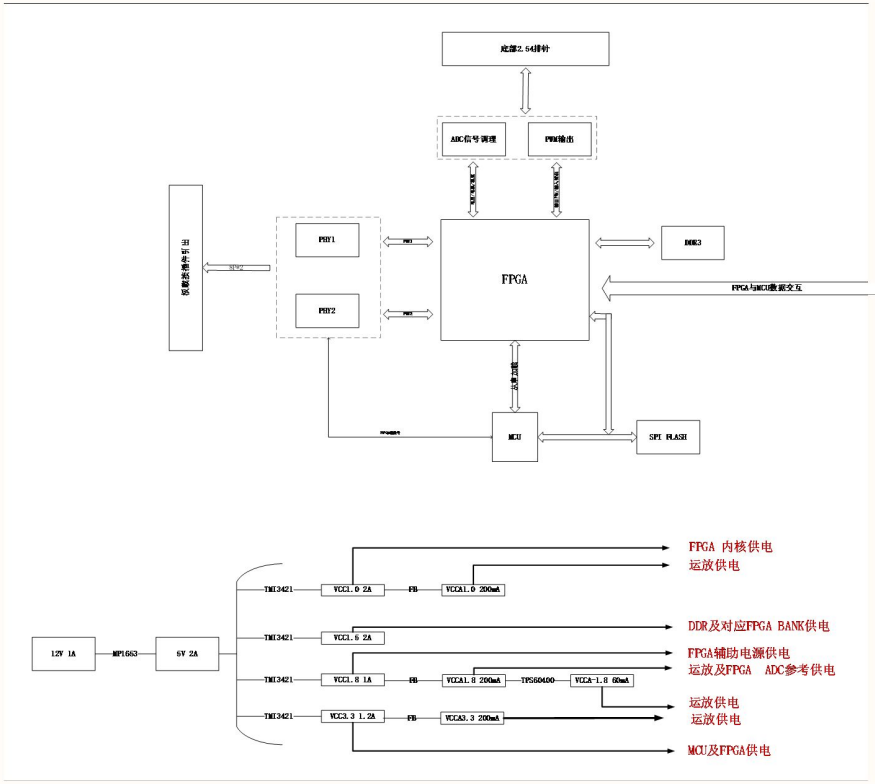
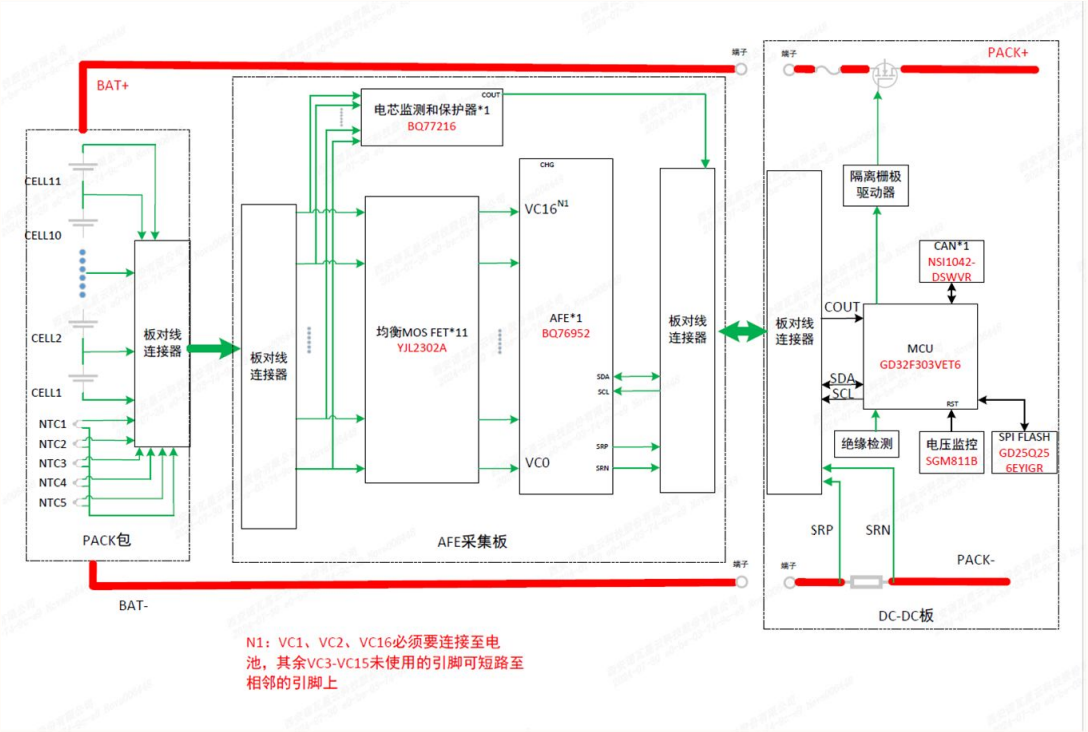




N1：VC1、VC2、VC16必须要连接至电池，其余VC3-VC15未使用的引脚可短路至相邻的引脚上

QR-CODE-500X500

MODEL

MESSAGE_BLOCK

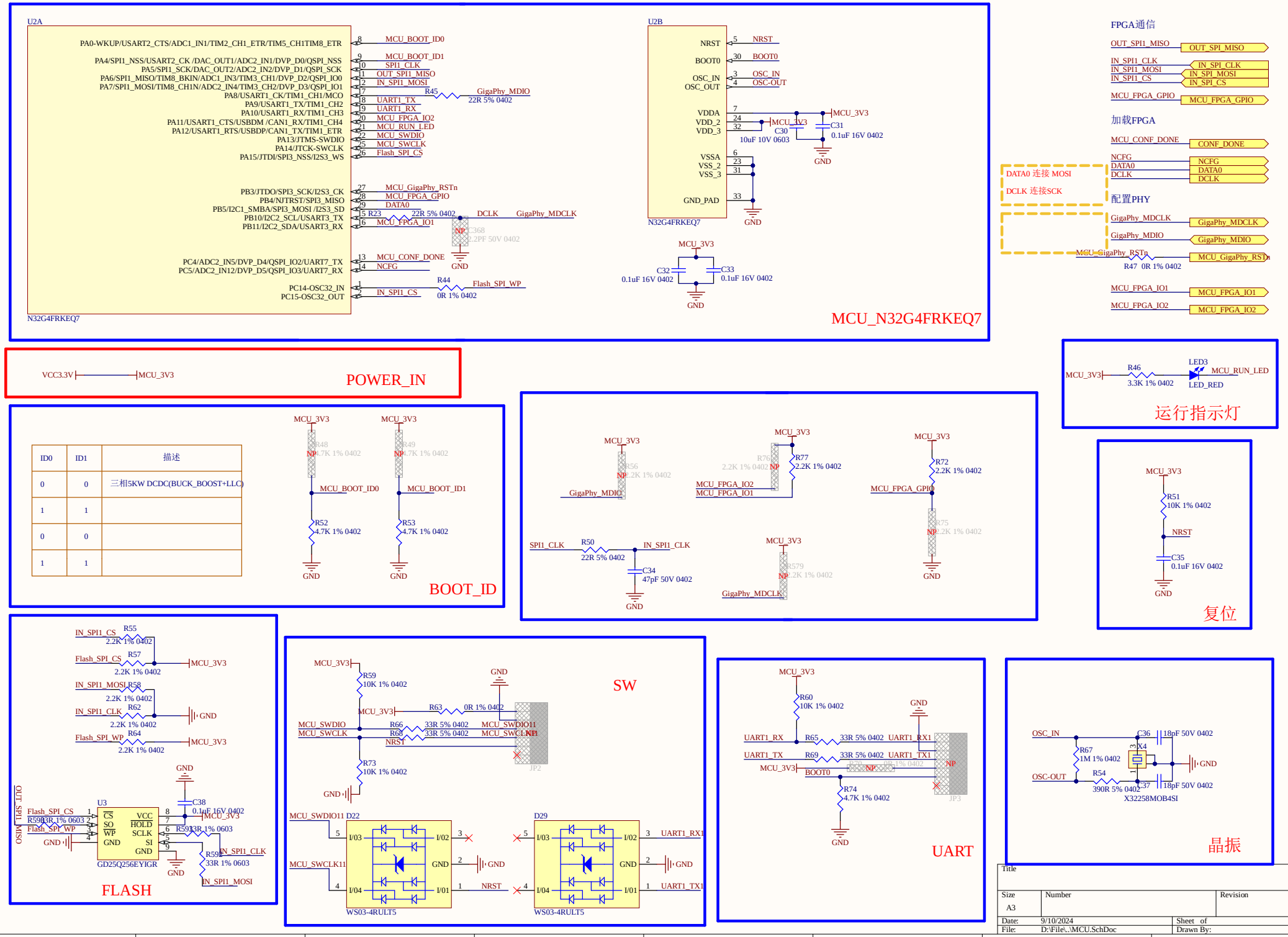
ESD_SILK

Date&QC

MARK2 Mark　MARK3 Mark　MARK6 Mark

MARK1 Mark　MARK4 Mark　MARK5 Mark

Title		
Size A1	Number	Revision
Date: 9/11/2024		Sheet of
File: D:\Har...\FOPCS.SchDoc		Drawn By:


U2A
PA0-WKUP/USART2_CTS/ADC1_IN1/TIM2_CH1_ETR/TIM5_CH1TIM8_ETR
PA4/SPI1_NSS/USART2_CK /DAC_OUT1/ADC2_IN1/DVP_D0/QSPI_NSS
PA5/SPI1_SCK/DAC_OUT2/ADC2_IN2/DVP_D1/QSPI_SCK
PA6/SPI1_MISO/TIM8_BKIN/ADC1_IN3/TIM3_CH1/DVP_D2/QSPI_IO0
PA7/SPI1_MOSI/TIM8_CH1N/ADC2_IN4/TIM3_CH2/DVP_D3/QSPI_IO1
PA8/USART1_CK/TIM1_CH1/MCO
PA9/USART1_TX/TIM1_CH2
PA10/USART1_RX/TIM1_CH3
PA11/USART1_CTS/USBDM /CAN1_RX/TIM1_CH4
PA12/USART1_RTS/USBDP/CAN1_TX/TIM1_ETR
PA13/JTMS-SWDIO
PA14/JTCK-SWCLK
PA15/JTDI/SPI3_NSS/I2S3_WS
PB3/JTDO/SPI3_SCK/I2S3_CK
PB4/NJTRST/SPI3_MISO
PB5/I2C1_SMBA/SPI3_MOSI /I2S3_SD
PB10/I2C2_SCL/USART3_TX
PB11/I2C2_SDA/USART3_RX
PC4/ADC2_IN5/DVP_D4/QSPI_IO2/UART7_TX
PC5/ADC2_IN12/DVP_D5/QSPI_IO3/UART7_RX
PC14-OSC32_IN
PC15-OSC32_OUT
MCU_BOOT_ID0
MCU_BOOT_ID1
SPI1_CLK
IN_SPI1_MOSI
OUT_SPI1_MISO
MCU_FPGA_IO2
MCU_RUN_LED
MCU_SWDIO
MCU_SWCLK
Flash_SPI_CS
UART1_TX
UART1_RX
MCU_GigaPhy_RSTn
MCU_FPGA_GPIO
DATA0
R23 22R 5% 0402
MCU_FPGA_IO1
DCLK
GigaPhy_MDCLK
NP C368 2.2PF 50V 0402
MCU_CONF_DONE
NCFG
R44 0R 1% 0402
Flash_SPI_WP
IN_SPI1_CS
R45 22R 5% 0402
GigaPhy_MDIO
N32G4FRKEQ7
U2B
NRST
BOOT0
OSC_IN
OSC_OUT
VDDA
VDD_2
VDD_3
VSSA
VSS_2
VSS_3
GND_PAD
MCU_3V3
C30 10uF 10V 0603
C31 0.1uF 16V 0402
C32 0.1uF 16V 0402
C33 0.1uF 16V 0402
GND
N32G4FRKEQ7
MCU_N32G4FRKEQ7
FPGA通信
OUT_SPI1_MISO
IN_SPI1_CLK
IN_SPI1_MOSI
IN_SPI1_CS
MCU_FPGA_GPIO
加载FPGA
MCU_CONF_DONE
CONF_DONE
NCFG
DATA0
DCLK
DATA0 连接 MOSI
DCLK 连接SCK
配置PHY
GigaPhy_MDCLK
GigaPhy_MDIO
MCU_GigaPhy_RSTn
R47 0R 1% 0402
MCU_FPGA_IO1
MCU_FPGA_IO2
VCC3.3V
MCU_3V3
POWER_IN
ID0
ID1
描述
0
0
三相5KW DCDC(BUCK_BOOST+LLC)
1
1
R48 NP 4.7K 1% 0402
R49 NP 4.7K 1% 0402
R52 4.7K 1% 0402
R53 4.7K 1% 0402
BOOT_ID
R56 NP 2.2K 1% 0402
R76 2.2K 1% 0402 NP
R77 2.2K 1% 0402
R72 2.2K 1% 0402
R75 NP 2.2K 1% 0402
R50 22R 5% 0402
C34 47pF 50V 0402
R79 NP 2.2K 1% 0402
SPI1_CLK
IN_SPI1_CLK
GigaPhy_MDCLK
LED3
R46 3.3K 1% 0402
LED_RED
运行指示灯
R51 10K 1% 0402
C35 0.1uF 16V 0402
复位
R55 2.2K 1% 0402
R57 2.2K 1% 0402
R58 2.2K 1% 0402
R62 2.2K 1% 0402
R64 2.2K 1% 0402
U3
C38 0.1uF 16V 0402
R593 3R 1% 0603
R592 33R 1% 0603
GD25Q256EYIGR
FLASH
R59 10K 1% 0402
R63 0R 1% 0402
R66 33R 5% 0402
R68 33R 5% 0402
MCU_SWDIO11
MCU_SWCLK11
NRST1
JP2
R73 10K 1% 0402
D22
D29
WS03-4RULT5
SW
R60 10K 1% 0402
R65 33R 5% 0402
R69 33R 5% 0402
UART1_RX1
UART1_TX1
R78 NP 1% 0402
JP3
R74 4.7K 1% 0402
UART
OSC_IN
OSC_OUT
R67 1M 1% 0402
R54 390R 5% 0402
C36 18pF 50V 0402
C37 18pF 50V 0402
X4
X32258MOB4SI
晶振
Title
Size
A3
Number
Revision
Date: 9/10/2024
File: D:\File\..\MCU.SchDoc
Sheet of
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POWER_IN

FPGA_BANK0

FPGA_配置模式

FPGA_XTAL_1

FPGA_运行晶振

FPGA_XTAL_2

FPGA_JTAG

FPGA_RUN_LED

FPGA&MCU

PHY-FPGA复位信号

DDR

使用FPGA BANK34

PHY

FPGA&BMS

PWM&IN

使用FPGA BANK16

FPGA_BANK_13

FPGA_BANK_14

FPGA_BANK_15

FPGA_BANK_16

DDR3_Vref使用电阻分压实现，因Vref电流较小，nmA

黄框组内数据DQ可以交换，DQS不能交换

控制地址信号可以交换

FPGA_BANK35

FPGA_BANK34

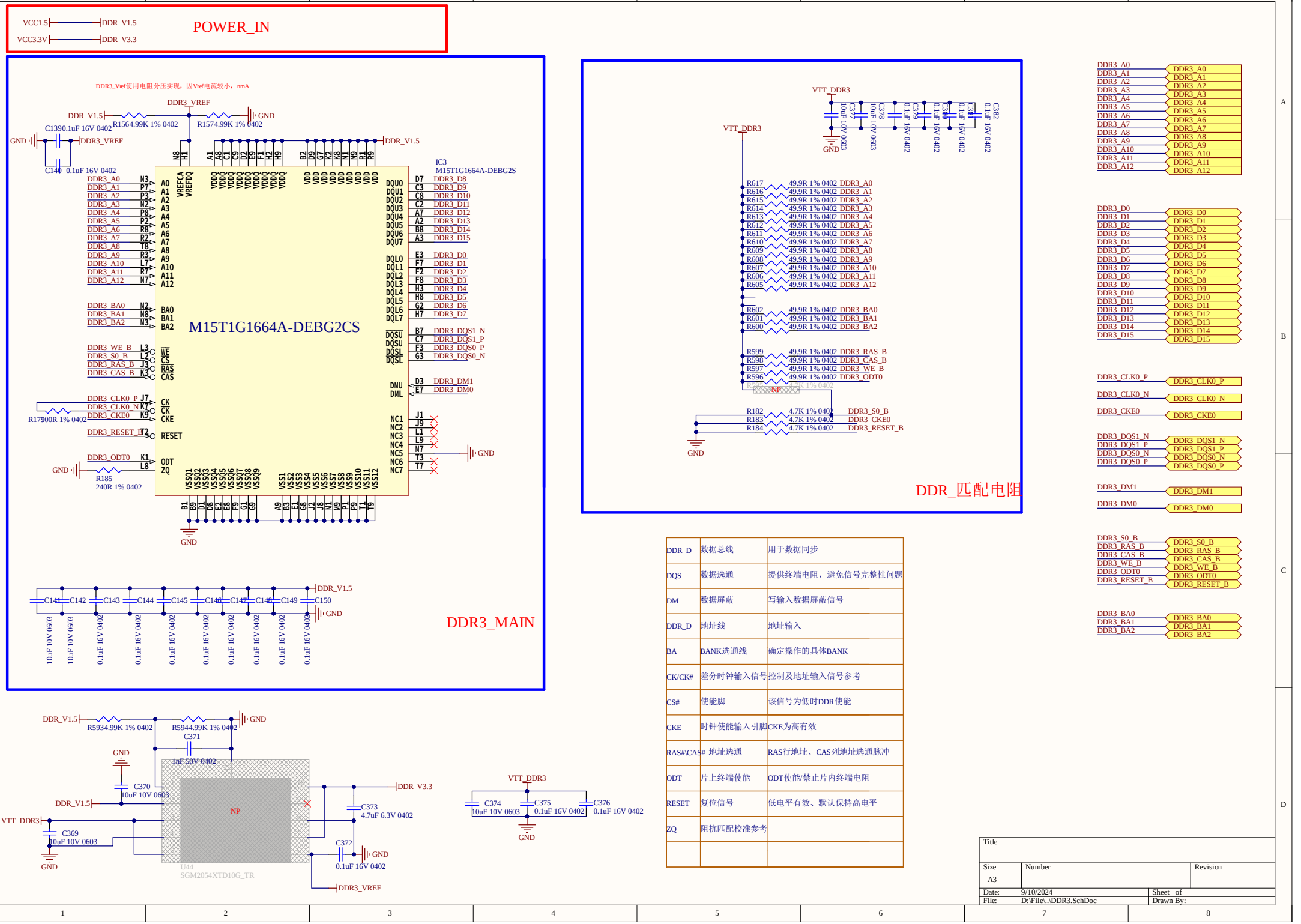
FPGA_BANK216

FPGA_POWER

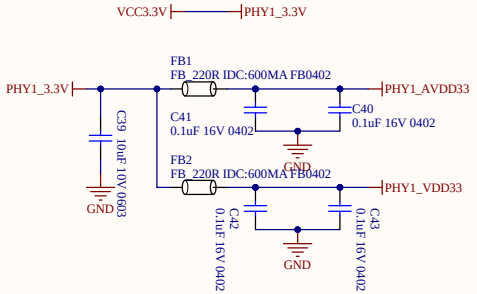
ADC

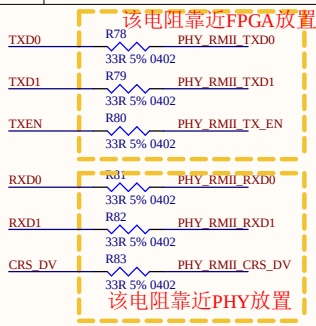
使用FPGA BANK15 BANK35

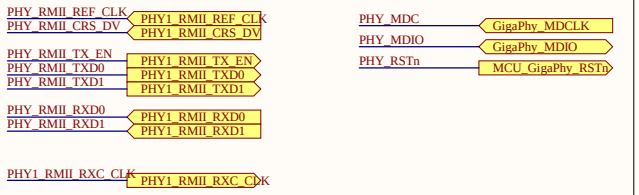
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Size A1	Number		Revision
Date: 9/11/2024		Sheet of	
File: D:\Fpga\...\FPGA3.SchDoc		Drawn By:	


POWER_IN
VCC1.5
DDR_V1.5
VCC3.3V
DDR_V3.3
DDR3_Vref使用电阻分压实现，因Vref电流较小，nmA
DDR3_VREF
R1564.99K 1% 0402
R1574.99K 1% 0402
C1390.1uF 16V 0402
C140 0.1uF 16V 0402
IC3
M15T1G1664A-DEBG2S
M15T1G1664A-DEBG2CS
R179 100R 1% 0402
R185 240R 1% 0402
DDR3_MAIN
R5934.99K 1% 0402
R5944.99K 1% 0402
C371 1nF 50V 0402
C370 10uF 10V 0603
C369 10uF 10V 0603
C373 4.7uF 6.3V 0402
C372 0.1uF 16V 0402
NP
U44
SGM2054XTD10G_TR
VTT_DDR3
C374 10uF 10V 0603
C375 0.1uF 16V 0402
C376 0.1uF 16V 0402
C377 10uF 10V 0603
C378 10uF 10V 0603
C379 0.1uF 16V 0402
C380 0.1uF 16V 0402
C381 0.1uF 16V 0402
C382 0.1uF 16V 0402
R617 49.9R 1% 0402 DDR3_A0
R616 49.9R 1% 0402 DDR3_A1
R615 49.9R 1% 0402 DDR3_A2
R614 49.9R 1% 0402 DDR3_A3
R613 49.9R 1% 0402 DDR3_A4
R612 49.9R 1% 0402 DDR3_A5
R611 49.9R 1% 0402 DDR3_A6
R610 49.9R 1% 0402 DDR3_A7
R609 49.9R 1% 0402 DDR3_A8
R608 49.9R 1% 0402 DDR3_A9
R607 49.9R 1% 0402 DDR3_A10
R606 49.9R 1% 0402 DDR3_A11
R605 49.9R 1% 0402 DDR3_A12
R602 49.9R 1% 0402 DDR3_BA0
R601 49.9R 1% 0402 DDR3_BA1
R600 49.9R 1% 0402 DDR3_BA2
R599 49.9R 1% 0402 DDR3_RAS_B
R598 49.9R 1% 0402 DDR3_CAS_B
R597 49.9R 1% 0402 DDR3_WE_B
R596 49.9R 1% 0402 DDR3_ODT0
R595 4.7K 1% 0402
R182 4.7K 1% 0402 DDR3_S0_B
R183 4.7K 1% 0402 DDR3_CKE0
R184 4.7K 1% 0402 DDR3_RESET_B
DDR_匹配电阻
DDR_D 数据总线 用于数据同步
DQS 数据选通 提供终端电阻，避免信号完整性问题
DM 数据屏蔽 写输入数据屏蔽信号
DDR_D 地址线 地址输入
BA BANK选通线 确定操作的具体BANK
CK/CK# 差分时钟输入信号 控制及地址输入信号参考
CS# 使能脚 该信号为低时DDR使能
CKE 时钟使能输入引脚 CKE为高有效
RAS#\CAS# 地址选通 RAS行地址、CAS列地址选通脉冲
ODT 片上终端使能 ODT使能/禁止片内终端电阻
RESET 复位信号 低电平有效、默认保持高电平
ZQ 阻抗匹配校准参考
Title
Size A3
Number
Revision
Date: 9/10/2024
Sheet of
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Drawn By:

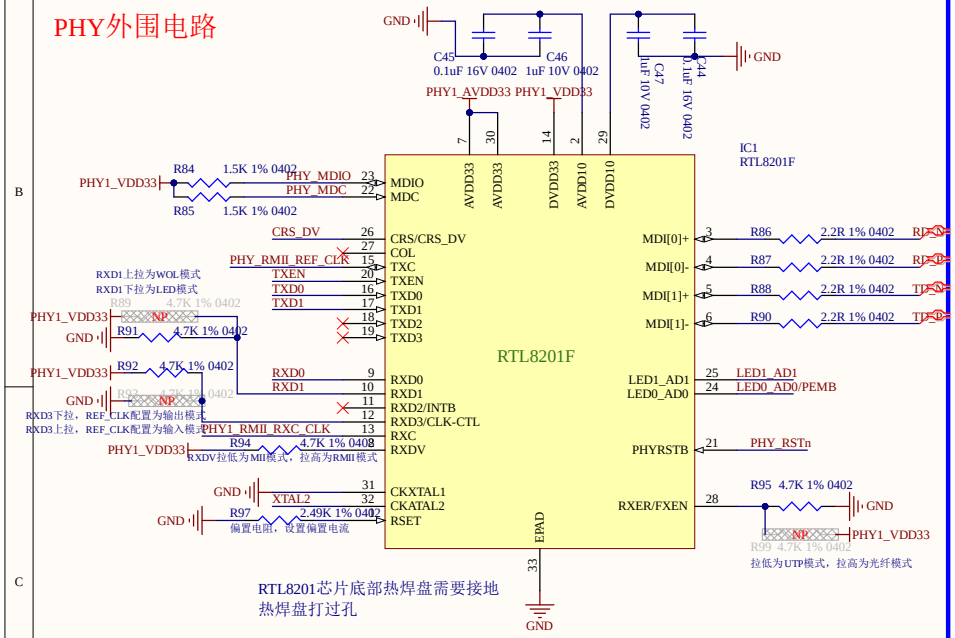
Power





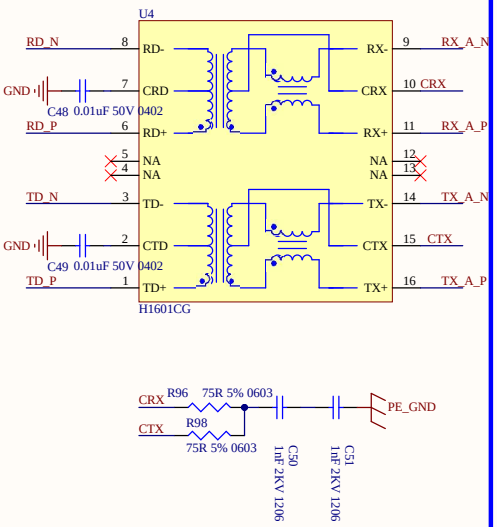


PHY外围电路

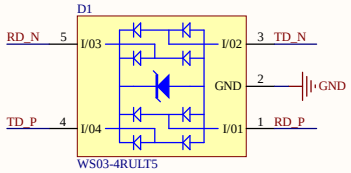


RTL8201芯片底部热焊盘需要接地
热焊盘打过孔

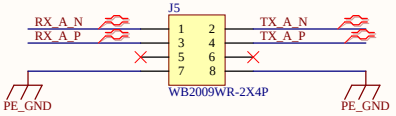
网络变压器

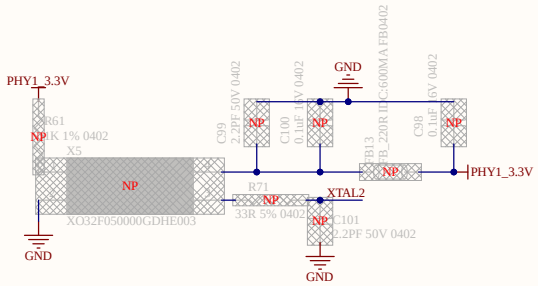


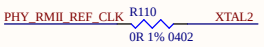
TVS



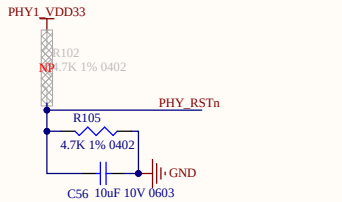
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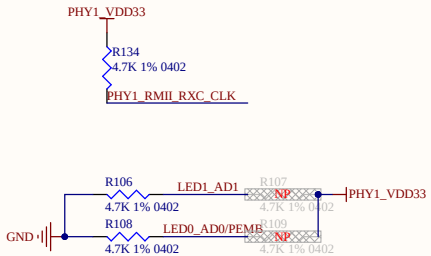




焊接RTL8201时，焊接R127,晶振电路NP，焊接YT8512取反



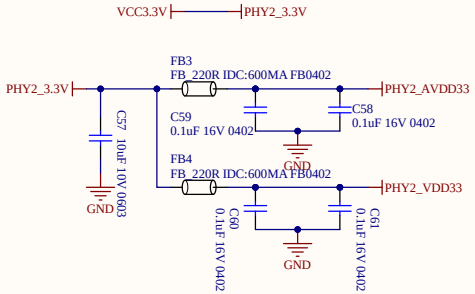
Note：1、在PHY的3.3V稳定之前复位为高至少80us；
2、要PHY完全复位，至少拉低需要10ms；
3、在PHY复位为高时至少150ms之后才可以访问PHY的寄存器。
4、复位与主板上PHY复位复用，因此，此处RC NP

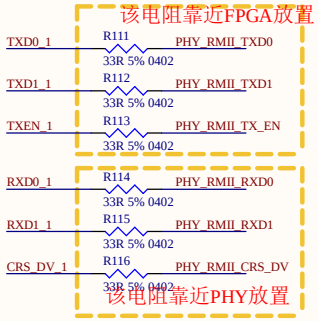


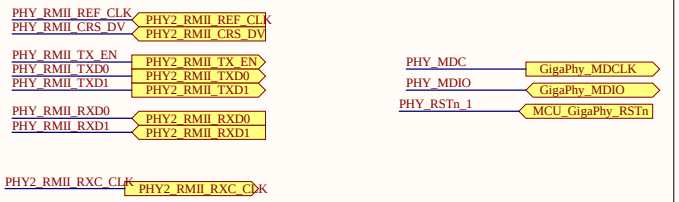
PHY地址配置为00

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Size A3	Number	Revision
Date: 9/10/2024		Sheet of
File: D:\File\..\PHY1.SchDoc		Drawn By:

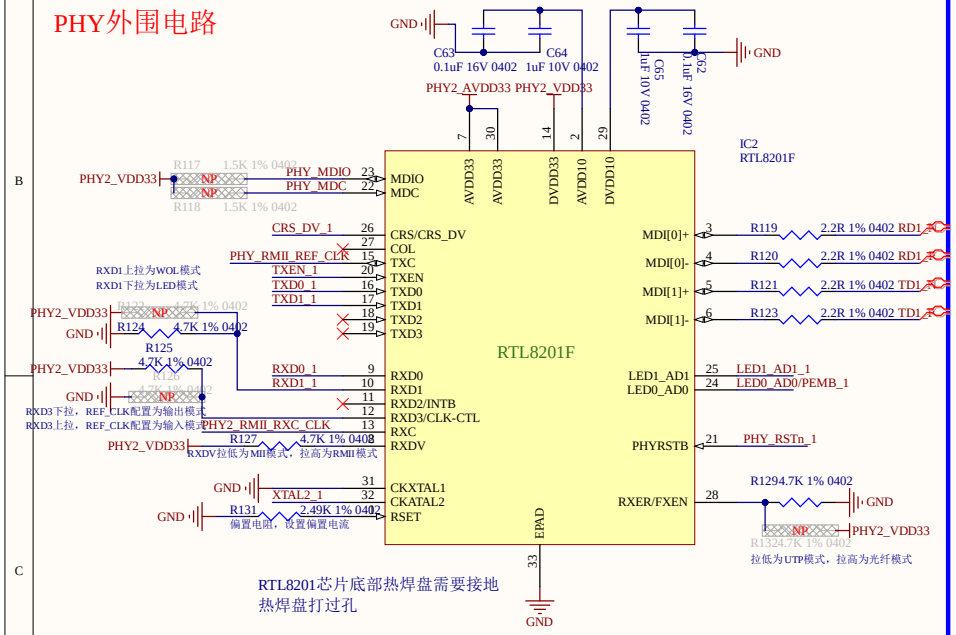
Power

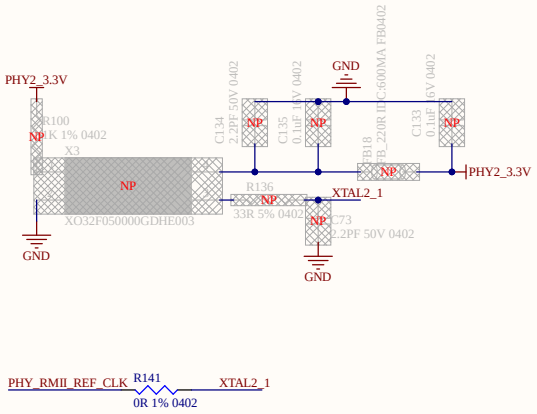




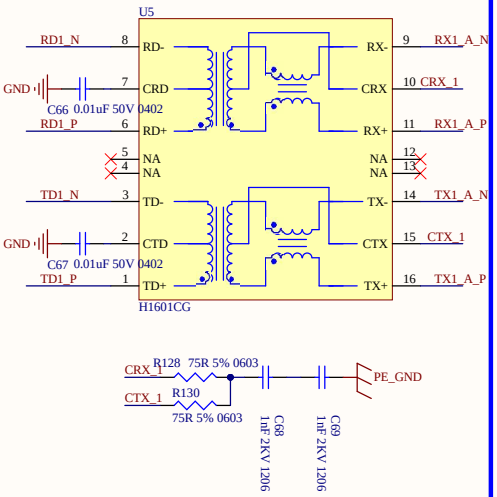


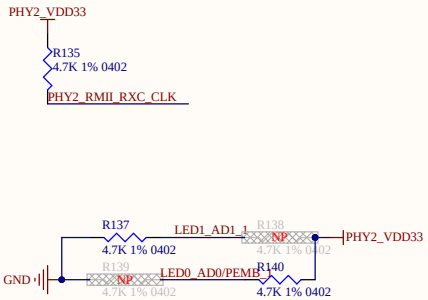
PHY外围电路



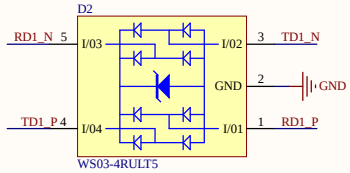


网络变压器

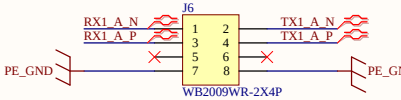




TVS



端子引出

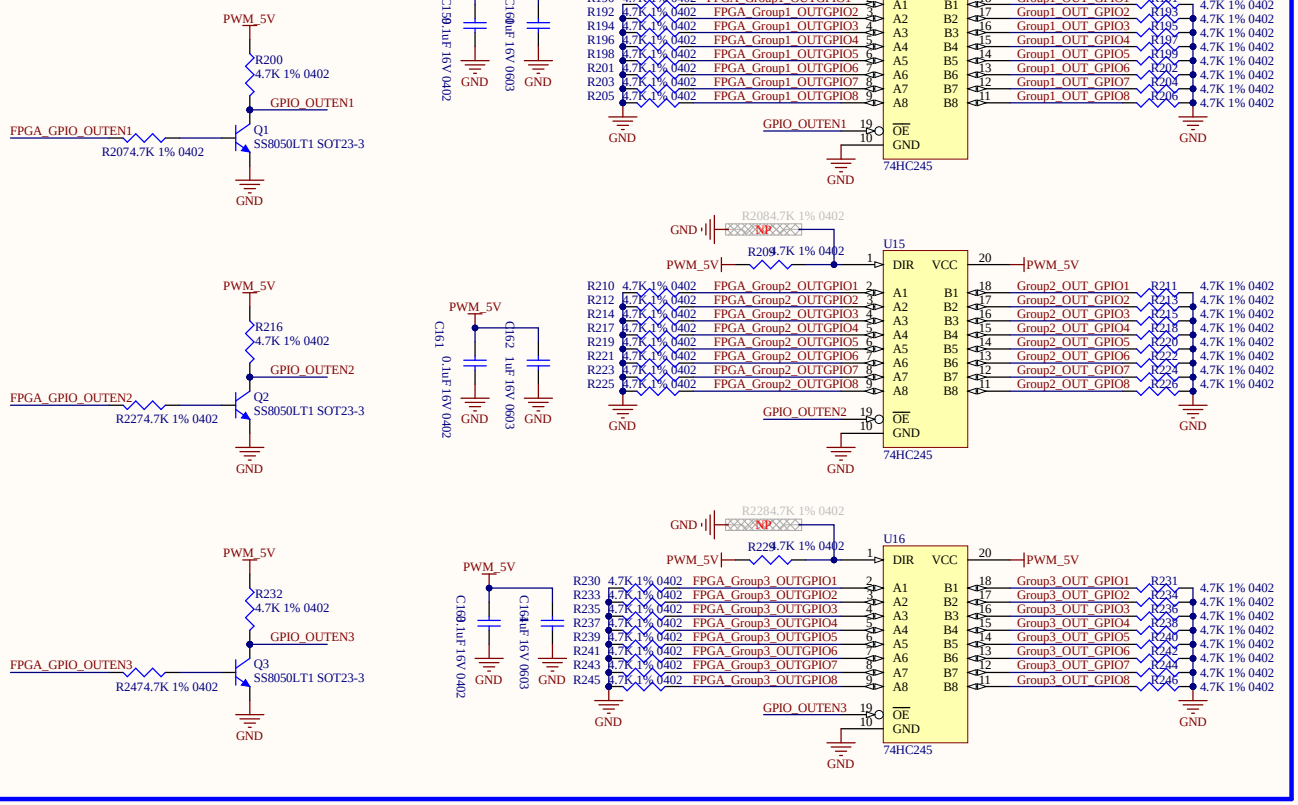


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Date: 9/10/2024		Sheet of
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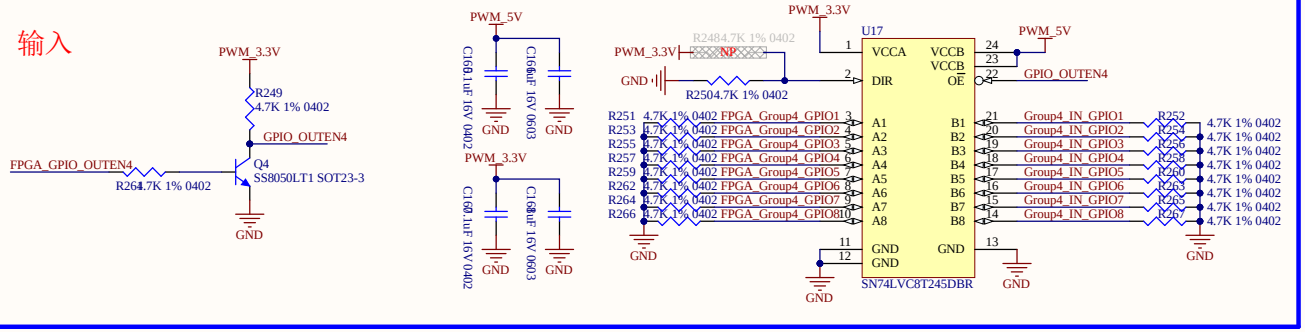
VCC5V — PWM_5V
VCC3.3V — PWM_3.3V

POWER_IN

输出



输入



FPGA_Group2_OUTGPIO1 — FPGA_Group2_OUTGPIO1
FPGA_Group2_OUTGPIO2 — FPGA_Group2_OUTGPIO2
FPGA_Group2_OUTGPIO3 — FPGA_Group2_OUTGPIO3
FPGA_Group2_OUTGPIO4 — FPGA_Group2_OUTGPIO4
FPGA_Group2_OUTGPIO5 — FPGA_Group2_OUTGPIO5
FPGA_Group2_OUTGPIO6 — FPGA_Group2_OUTGPIO6
FPGA_Group2_OUTGPIO7 — FPGA_Group2_OUTGPIO7
FPGA_Group2_OUTGPIO8 — FPGA_Group2_OUTGPIO8

FPGA_Group1_OUTGPIO1 — FPGA_Group1_OUTGPIO1
FPGA_Group1_OUTGPIO2 — FPGA_Group1_OUTGPIO2
FPGA_Group1_OUTGPIO3 — FPGA_Group1_OUTGPIO3
FPGA_Group1_OUTGPIO4 — FPGA_Group1_OUTGPIO4
FPGA_Group1_OUTGPIO5 — FPGA_Group1_OUTGPIO5
FPGA_Group1_OUTGPIO6 — FPGA_Group1_OUTGPIO6
FPGA_Group1_OUTGPIO7 — FPGA_Group1_OUTGPIO7
FPGA_Group1_OUTGPIO8 — FPGA_Group1_OUTGPIO8

FPGA_Group4_GPIO1 — FPGA_Group4_GPIO1
FPGA_Group4_GPIO2 — FPGA_Group4_GPIO2
FPGA_Group4_GPIO3 — FPGA_Group4_GPIO3
FPGA_Group4_GPIO4 — FPGA_Group4_GPIO4
FPGA_Group4_GPIO5 — FPGA_Group4_GPIO5
FPGA_Group4_GPIO6 — FPGA_Group4_GPIO6
FPGA_Group4_GPIO7 — FPGA_Group4_GPIO7
FPGA_Group4_GPIO8 — FPGA_Group4_GPIO8

FPGA_Group3_OUTGPIO1 — FPGA_Group3_OUTGPIO1
FPGA_Group3_OUTGPIO2 — FPGA_Group3_OUTGPIO2
FPGA_Group3_OUTGPIO3 — FPGA_Group3_OUTGPIO3
FPGA_Group3_OUTGPIO4 — FPGA_Group3_OUTGPIO4
FPGA_Group3_OUTGPIO5 — FPGA_Group3_OUTGPIO5
FPGA_Group3_OUTGPIO6 — FPGA_Group3_OUTGPIO6
FPGA_Group3_OUTGPIO7 — FPGA_Group3_OUTGPIO7
FPGA_Group3_OUTGPIO8 — FPGA_Group3_OUTGPIO8

FPGA_GPIO_OUTEN1 — FPGA_GPIO_OUTEN1
FPGA_GPIO_OUTEN2 — FPGA_GPIO_OUTEN2
FPGA_GPIO_OUTEN3 — FPGA_GPIO_OUTEN3
FPGA_GPIO_OUTEN4 — FPGA_GPIO_OUTEN4

Group2_OUT_GPIO1 — Group2_OUTGPIO1
Group2_OUT_GPIO2 — Group2_OUTGPIO2
Group2_OUT_GPIO3 — Group2_OUTGPIO3
Group2_OUT_GPIO4 — Group2_OUTGPIO4
Group2_OUT_GPIO5 — Group2_OUTGPIO5
Group2_OUT_GPIO6 — Group2_OUTGPIO6
Group2_OUT_GPIO7 — Group2_OUTGPIO7
Group2_OUT_GPIO8 — Group2_OUTGPIO8

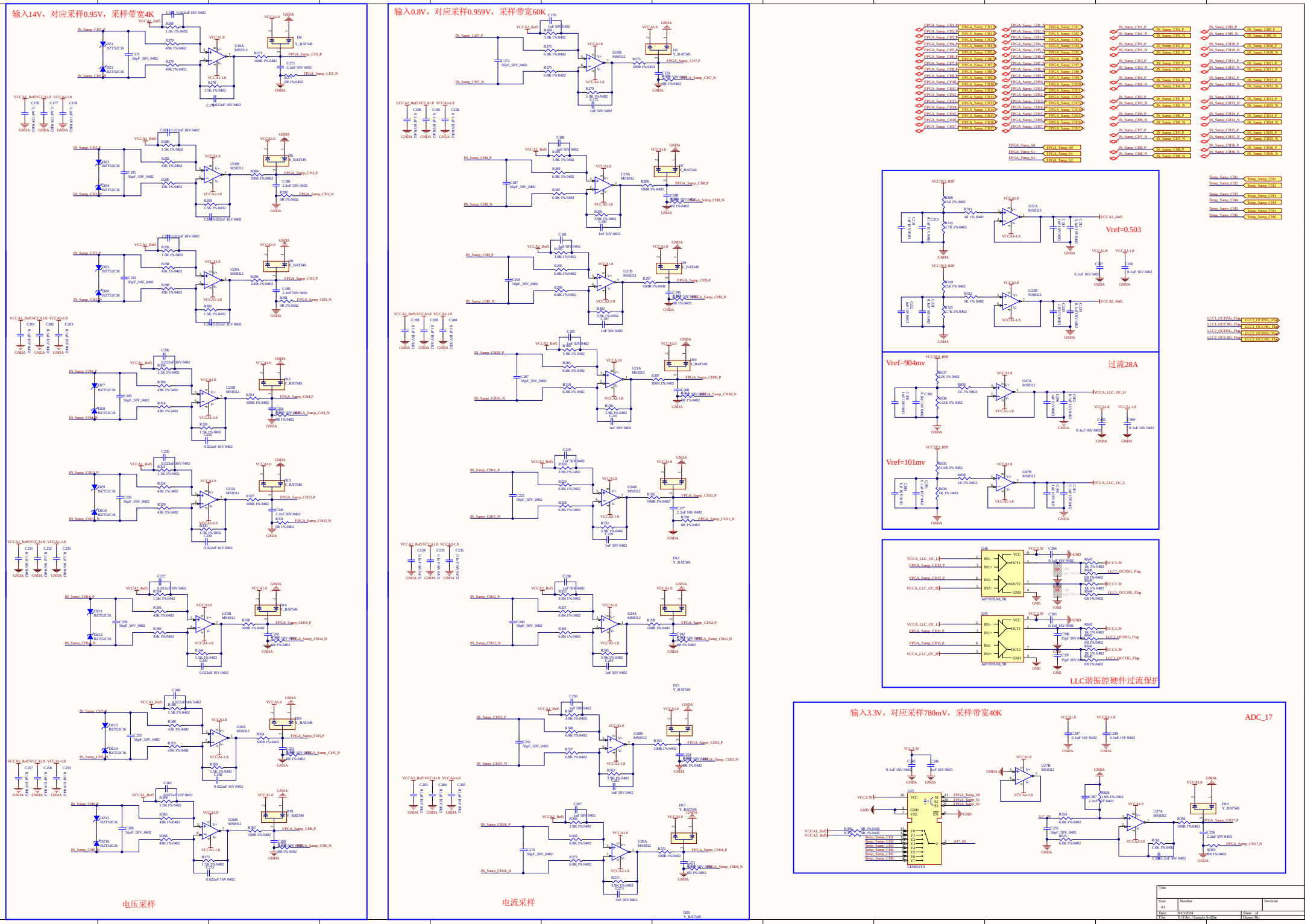
Group1_OUT_GPIO1 — Group1_OUTGPIO1
Group1_OUT_GPIO2 — Group1_OUTGPIO2
Group1_OUT_GPIO3 — Group1_OUTGPIO3
Group1_OUT_GPIO4 — Group1_OUTGPIO4
Group1_OUT_GPIO5 — Group1_OUTGPIO5
Group1_OUT_GPIO6 — Group1_OUTGPIO6
Group1_OUT_GPIO7 — Group1_OUTGPIO7
Group1_OUT_GPIO8 — Group1_OUTGPIO8

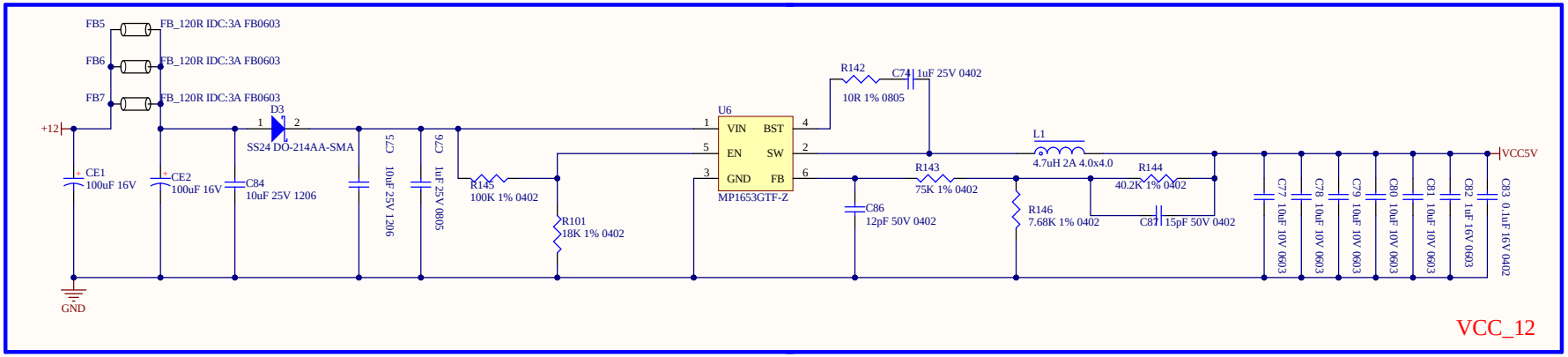
Group4_IN_GPIO1 — Group4_GPIO1
Group4_IN_GPIO2 — Group4_GPIO2
Group4_IN_GPIO3 — Group4_GPIO3
Group4_IN_GPIO4 — Group4_GPIO4
Group4_IN_GPIO5 — Group4_GPIO5
Group4_IN_GPIO6 — Group4_GPIO6
Group4_IN_GPIO7 — Group4_GPIO7
Group4_IN_GPIO8 — Group4_GPIO8

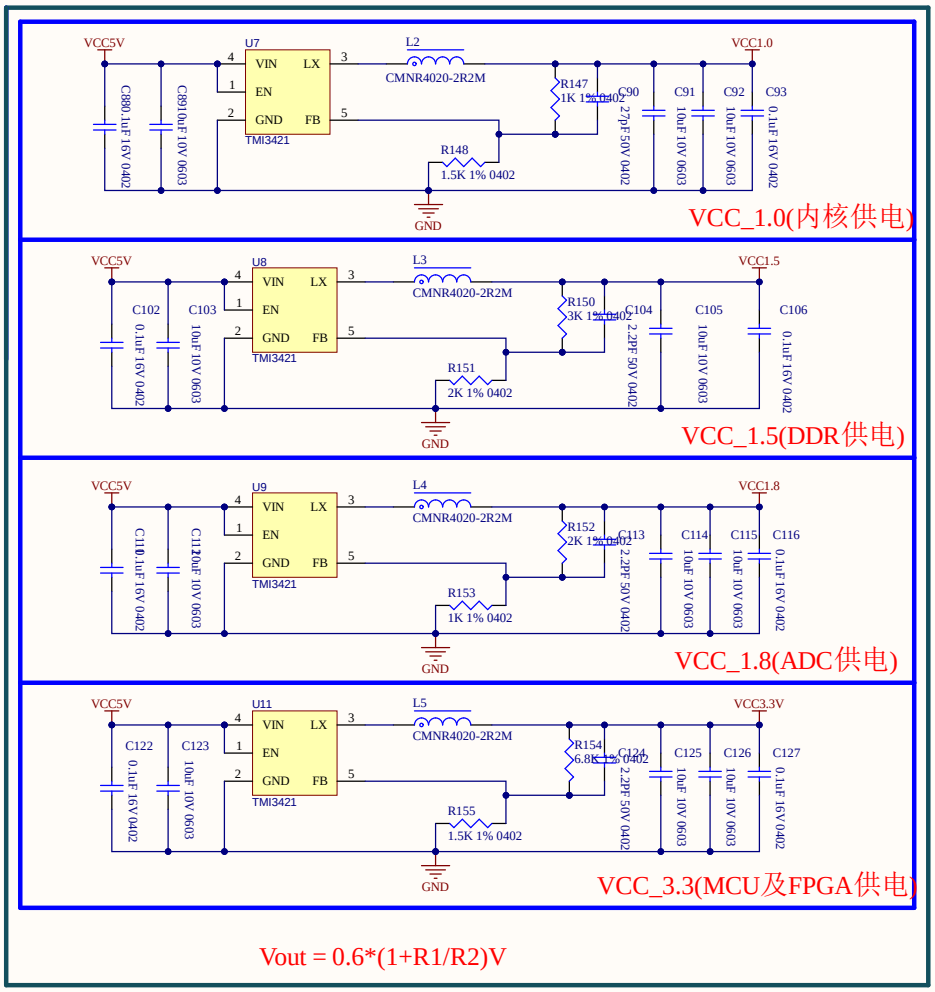
Group3_OUT_GPIO1 — Group3_OUTGPIO1
Group3_OUT_GPIO2 — Group3_OUTGPIO2
Group3_OUT_GPIO3 — Group3_OUTGPIO3
Group3_OUT_GPIO4 — Group3_OUTGPIO4
Group3_OUT_GPIO5 — Group3_OUTGPIO5
Group3_OUT_GPIO6 — Group3_OUTGPIO6
Group3_OUT_GPIO7 — Group3_OUTGPIO7
Group3_OUT_GPIO8 — Group3_OUTGPIO8

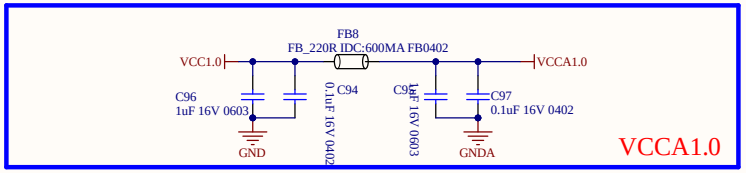
拉高使能PWM输出，拉低之后PWM为0
245之前下拉是为了加载程序的时候防止FPGA输出不定态
245之后下拉是为了刹车信号来的时候，给到驱动板的是低电平

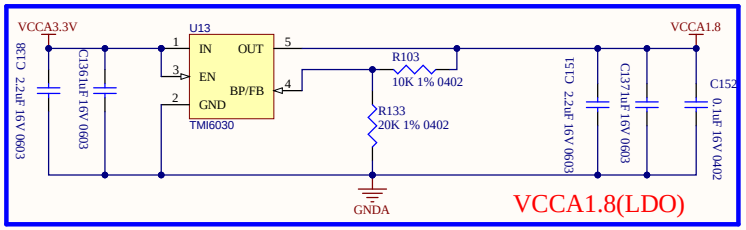
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Size A3	Number	Revision
Date: 9/10/2024		Sheet of
File: D:\File\..\OUT_INCtrl.SchDoc		Drawn By:

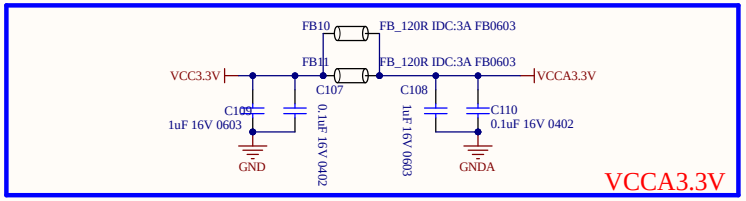

输入14V，对应采样0.95V，采样带宽4K
电压采样
输入0.8V，对应采样0.959V，采样带宽60K
电流采样
Vref=0.503
Vref=904mv
过流28A
Vref=101mv
LLC谐振腔硬件过流保护
输入3.3V，对应采样780mV，采样带宽40K
ADC_17
Title
Size
A1
Number
Revision
Date:
File:
Sheet of
Drawn By:

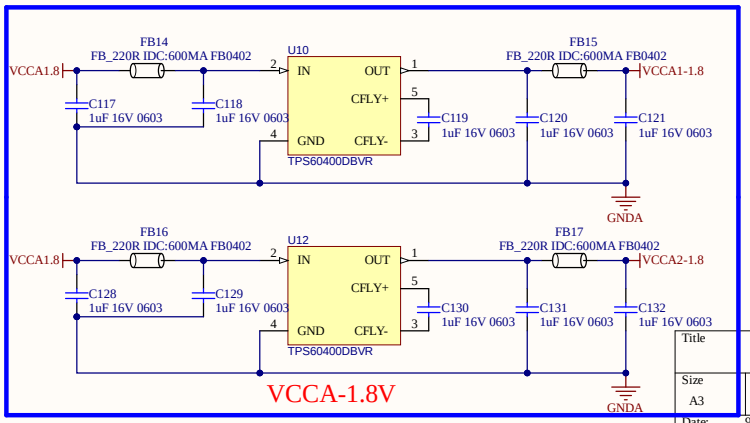


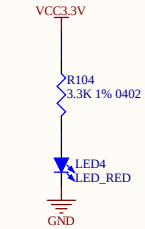








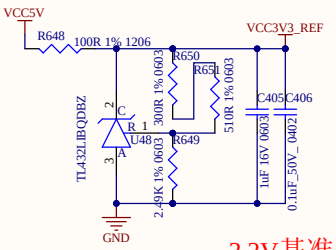




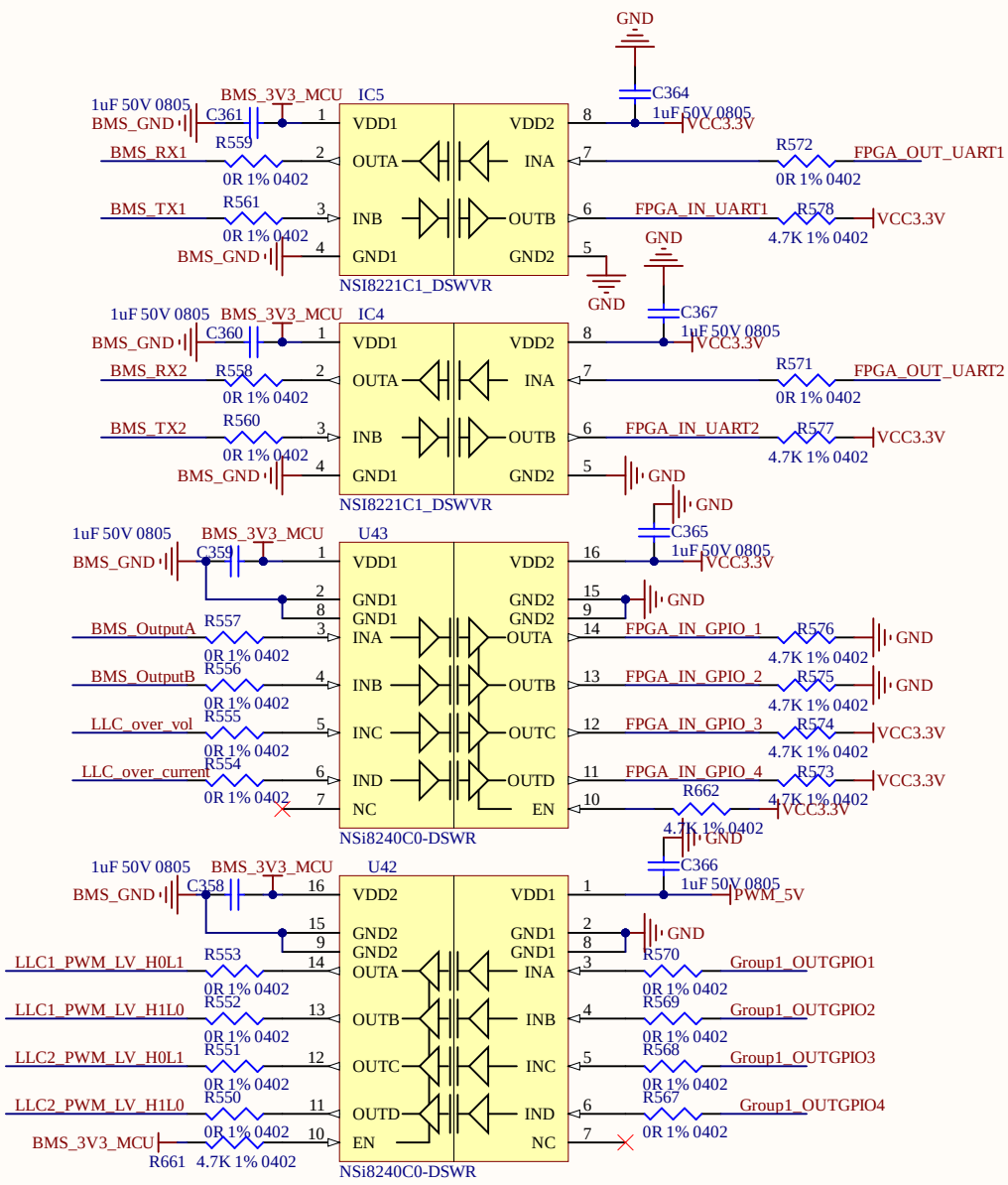


模拟数字地（单点接地）

$$3V3_REF = \frac{2.495\times(2.49K+810)}{2.49K} = 3.3066V$$



Title		
Size: A3	Number	Revision
Date: 9/10/2024		Sheet of
File: D:\File\..\POWER.SchDoc		Drawn By:



LLC1_HV_PWM_H1L0
LLC1_HV_PWM_H0L1
LLC2_HV_PWM_H1L0
LLC2_HV_PWM_H0L1

Group1_OUTGPIO1 — Group1_OUTGPIO1
Group1_OUTGPIO2 — Group1_OUTGPIO2
Group1_OUTGPIO3 — Group1_OUTGPIO3
Group1_OUTGPIO4 — Group1_OUTGPIO4

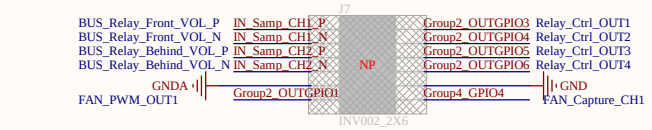
FPGA_IN_GPIO_1 — FPGA_IN_GPIO_1
FPGA_IN_GPIO_2 — FPGA_IN_GPIO_2
FPGA_IN_GPIO_3 — FPGA_IN_GPIO_3
FPGA_IN_GPIO_4 — FPGA_IN_GPIO_4

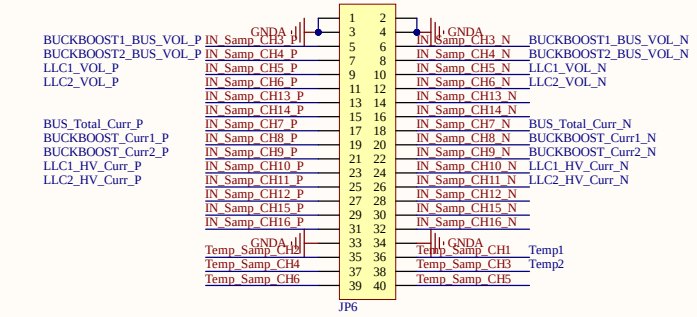
FPGA_IN_UART1 — FPGA_IN_UART1
FPGA_IN_UART2 — FPGA_IN_UART2
FPGA_OUT_UART1 — FPGA_OUT_UART1
FPGA_OUT_UART2 — FPGA_OUT_UART2

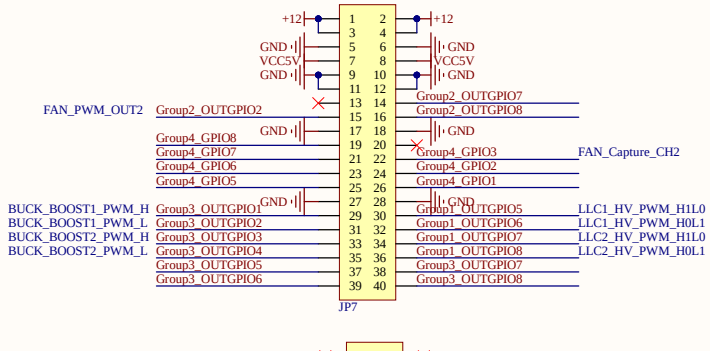
TO BMS

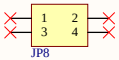
BMS_RX2 — BMS_RX2
BMS_TX2 — BMS_TX2
BMS_RX1 — BMS_RX1
BMS_TX1 — BMS_TX1
BMS_OutputA — BMS_OutputA
BMS_OutputB — BMS_OutputB
LLC_over_vol — LLC_over_vol
LLC_over_current — LLC_over_current
LLC1_PWM_LV_H0L1 — LLC1_PWM_LV_H0L1
LLC1_PWM_LV_H1L0 — LLC1_PWM_LV_H1L0
LLC2_PWM_LV_H0L1 — LLC2_PWM_LV_H0L1
LLC2_PWM_LV_H1L0 — LLC2_PWM_LV_H1L0

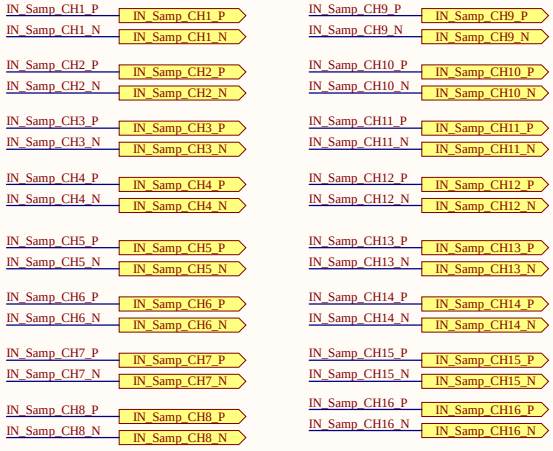
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Date: 9/10/2024		Sheet of
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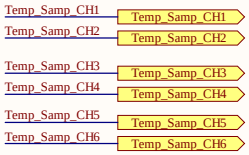


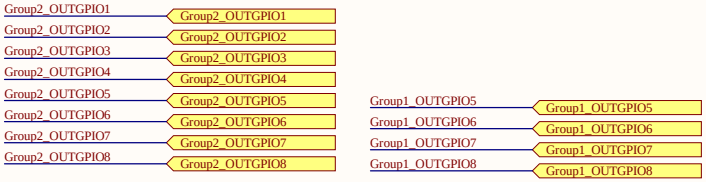


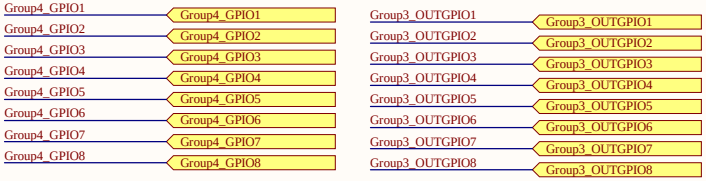




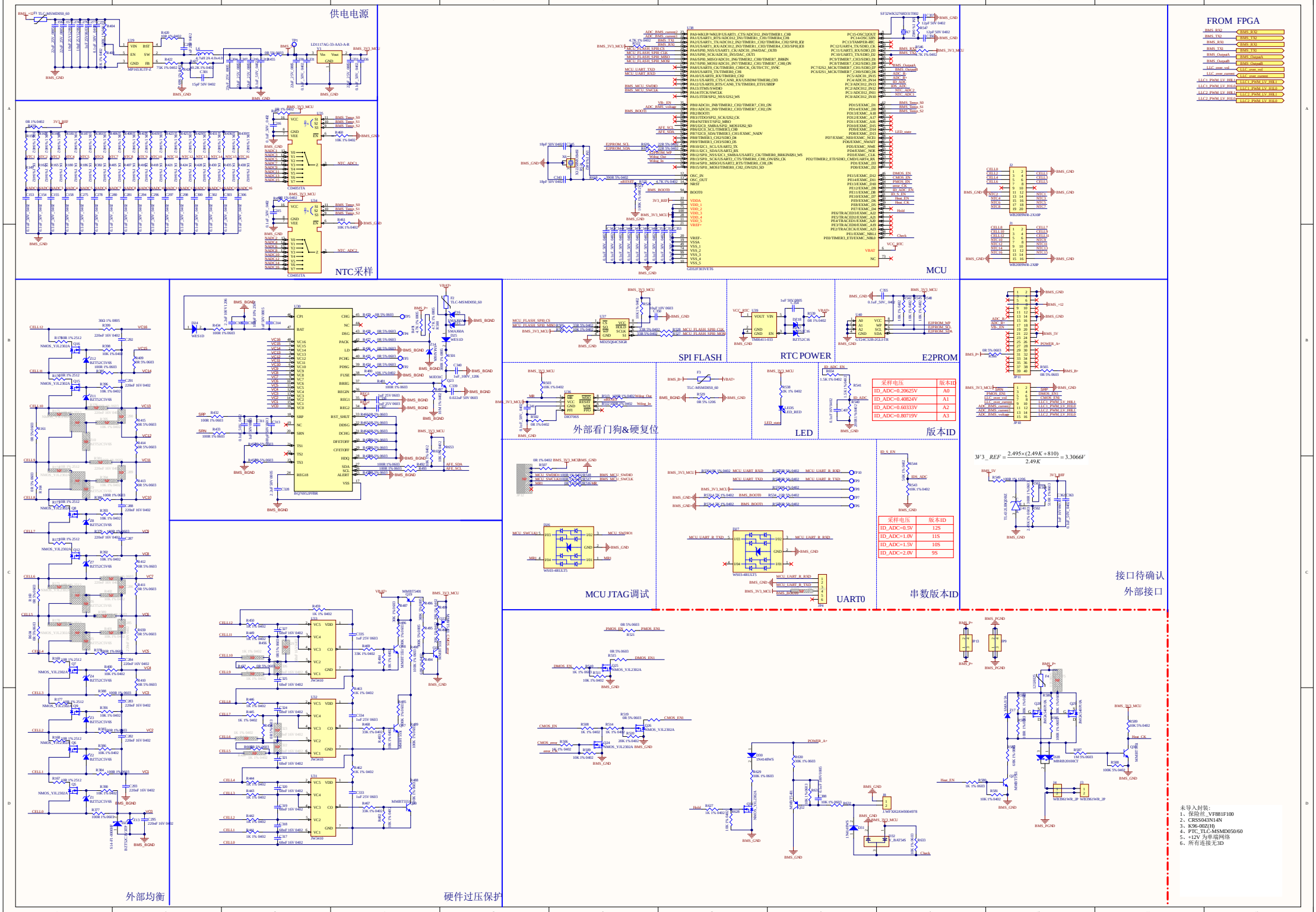








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Size: A3	Number	Revision
Date: 9/10/2024		Sheet of
File: D:\File\..\Interface.SchDoc		Drawn By:


供电电源
NTC采样
MCU
FROM FPGA
SPI FLASH
RTC POWER
E2PROM
外部看门狗&硬复位
LED
版本ID
MCU JTAG调试
UART0
串数版本ID
接口待确认
外部接口
外部均衡
硬件过压保护
采样电压 版本ID
ID_ADC=0.20625V A0
ID_ADC=0.40824V A1
ID_ADC=0.60333V A2
ID_ADC=0.80719V A3
采样电压 版本ID
ID_ADC=0.5V 12S
ID_ADC=1.0V 11S
ID_ADC=1.5V 10S
ID_ADC=2.0V 9S
未导入封装：
1、保险丝_VFB81F100
2、CRSS043N14N
3、K96-002(H)
4、PTC_TLC-MSMD050/60
5、+12V 为单端网络
6、所有连接无3D